

A Project Report on
Data Visualization With Power BI



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Acknowledgement

I sincerely acknowledge the guidance and support received during the course of this project. The availability of reliable datasets and the use of Microsoft Power BI made it possible to transform raw information into meaningful insights. I am grateful to my peers and mentors who encouraged me to explore advanced visualization techniques and to the gaming industry reports that provided context for interpreting the data. Their contributions collectively shaped the success of this project.

Introduction

The video game industry has evolved into one of the largest entertainment sectors globally, with PC gaming playing a pivotal role. Over the past decade, blockbuster titles such as *PUBG*, *Elden Ring*, and *Baldur's Gate 3* have not only captivated audiences but also generated billions in revenue. This project focuses on analyzing PC game performance between 2015 and 2024, covering aspects such as sales, revenue, developer distribution, and ratings. By creating a comprehensive dashboard, the aim is to provide a clear picture of industry trends and highlight the factors that drive success in the gaming market.

Objectives of the Project

The project was designed with the following objectives:

1. To study **sales and revenue trends** of PC games over a decade.
2. To identify **top-performing titles** and their contribution to industry growth.
3. To analyze **developer distribution across digital platforms** such as Steam, Epic, GOG, and Rockstar.
4. To evaluate **IGN ratings trends** and their impact on player engagement.
5. To design an **interactive dashboard** that simplifies complex datasets into actionable insights for decision-makers.

Tools and Techniques Used

- **Microsoft Power BI:** Used for building interactive dashboards and visualizations.
- **Excel/CSV:** For dataset preparation, cleaning, and formatting.
- **Data Modeling:** Establishing relationships between attributes like year, title, and developer.
- **Visualization Techniques:** Employing bar charts for comparisons, line charts for trends, pie charts for proportions, and KPIs for quick highlights.
- **Analytical Methods:** Trend analysis, comparative analysis, and correlation between ratings and sales.

Dataset Description

The dataset spans 2015–2024 and includes:

- **Year of Release** – to analyze chronological trends.
- **Title of Game** – popular PC titles across genres.
- **Genre** – RPG, Action, Survival, Shooter, etc.
- **Developer** – studios like CD Projekt Red, Larian Studios, FromSoftware, and Capcom.
- **PC Units Sold (Millions)** – sales performance.
- **Revenue (Millions)** – financial success.
- **IGN Ratings** – critical reception and player reviews.
- **Digital Store Distribution** – platforms such as Steam, Epic, GOG, MS Store, and Rockstar.

This dataset provides a holistic view of the industry, combining commercial, critical, and distribution aspects.

Data Cleaning Process

Data cleaning was a crucial step to ensure accuracy:

- **Duplicate Removal:** Eliminated repeated entries for the same game across years.
- **Standardization:** Unified naming conventions for developers and digital stores.
- **Handling Missing Values:** Filled gaps in ratings and sales using averages or reliable estimates.
- **Numeric Formatting:** Converted revenue and sales into consistent million-dollar/million-unit formats.
- **Genre Classification:** Ensured uniform categorization (e.g., “Action RPG” vs “RPG”).

This process improved dataset reliability and made the dashboard more trustworthy.

Overview of the Dashboard

The dashboard provides a **comprehensive snapshot** of the PC gaming industry:

- **KPIs:** Quick highlights of top units sold (25M) and top grossing game (\$900M).
- **Bar Charts:** Comparative analysis of sales and revenue across major titles.
- **Pie Chart:** Developer distribution across digital platforms.
- **Line Charts:** Trends in sales, revenue, and ratings from 2016–2024.

Together, these visuals allow users to explore both micro-level (individual games) and macro-level (industry trends) insights.

Visuals Used & Key Findings

- **Bar Charts:** Showed that *PUBG* dominated unit sales while *Baldur's Gate 3* led in revenue.
- **Pie Chart:** Revealed Steam's dominance with nearly 39% developer share.
- **Line Charts:** Illustrated steady growth in sales and revenue post-2017, with peaks in 2023–2024.
- **KPIs:** Highlighted quick insights for decision-makers.

Key Findings:

- *PUBG* achieved the highest unit sales (25M).
- *Baldur's Gate 3* was the top-grossing game (\$900M).
- Steam remains the most preferred platform for developers.
- IGN ratings peaked with *Baldur's Gate 3* (96), showing critical acclaim aligns with revenue success.
- The industry saw consistent growth, with 2023–2024 marking a golden period for PC gaming

Features of the Dashboard

- **Interactive Filters:** Year, genre, and developer-based exploration.
- **Combined KPIs and Charts:** Quick highlights alongside detailed analysis.
- **Trend Analysis:** Yearly progression of sales, revenue, and ratings.
- **Comparative Insights:** Ability to compare games side by side.
- **User-Friendly Layout:** Professional design with clear legends, values, and labels.

Conclusion

The PC Games Dashboard project successfully demonstrates how data visualization can transform raw datasets into actionable insights. By analyzing sales, revenue, ratings, and developer distribution, the dashboard provides a holistic view of the gaming industry. The findings highlight the dominance of certain titles, the importance of digital platforms, and the correlation between critical ratings and commercial success. This project not only strengthens analytical and technical skills but also offers valuable perspectives for developers, publishers, and analysts in planning future strategies.

Appendix

- **Dataset Sources:** Game sales, revenue, developer distribution, and IGN ratings (2016–2024).
- **Tools Used:** Microsoft Power BI for visualization, Excel/CSV for dataset preparation.
- **Charts Included:**
 - Bar charts for unit sales and revenue by title
 - Pie chart for developer distribution across digital stores
 - Line charts for yearly sales, revenue, and ratings trends
- **Key Metrics:**
 - Top PC Units Sold: 25M
 - Top Grossing Game: \$900M
- **References**
 - Microsoft Power BI Documentation
 - IGN Ratings Database
 - Steam, Epic Games, GOG, Rockstar, and MS Store official data
 - Industry reports on PC gaming trends (2015–2024)
 - Developer websites and press releases

Future Scope

- **Integration of Real-Time Data:** Connect APIs from Steam, Epic, and other stores for live updates.
- **Predictive Analytics:** Use machine learning models to forecast future sales and ratings.
- **Expanded Metrics:** Include player engagement, in-game purchases, and regional sales distribution.
- **Cross-Platform Comparison:** Extend dashboard to compare PC vs console vs mobile gaming trends.
- **Interactive Features:** Add filters for genre, developer, and release year to enhance user exploration.

Learning from the Project

- **Data Cleaning Importance:** Handling missing values and ensuring consistency across multiple sources was crucial.
- **Visualization Choices:** Selecting the right chart type (bar for comparisons, line for trends, pie for proportions) improved clarity.
- **Insight Generation:** The dashboard highlighted how certain titles dominate both sales and revenue, while developer distribution shows platform preferences.
- **Technical Skills:** Strengthened knowledge of Power BI, data modeling, and dashboard storytelling.
- **Analytical Thinking:** Learned to connect raw data with meaningful business insights for decision-making.